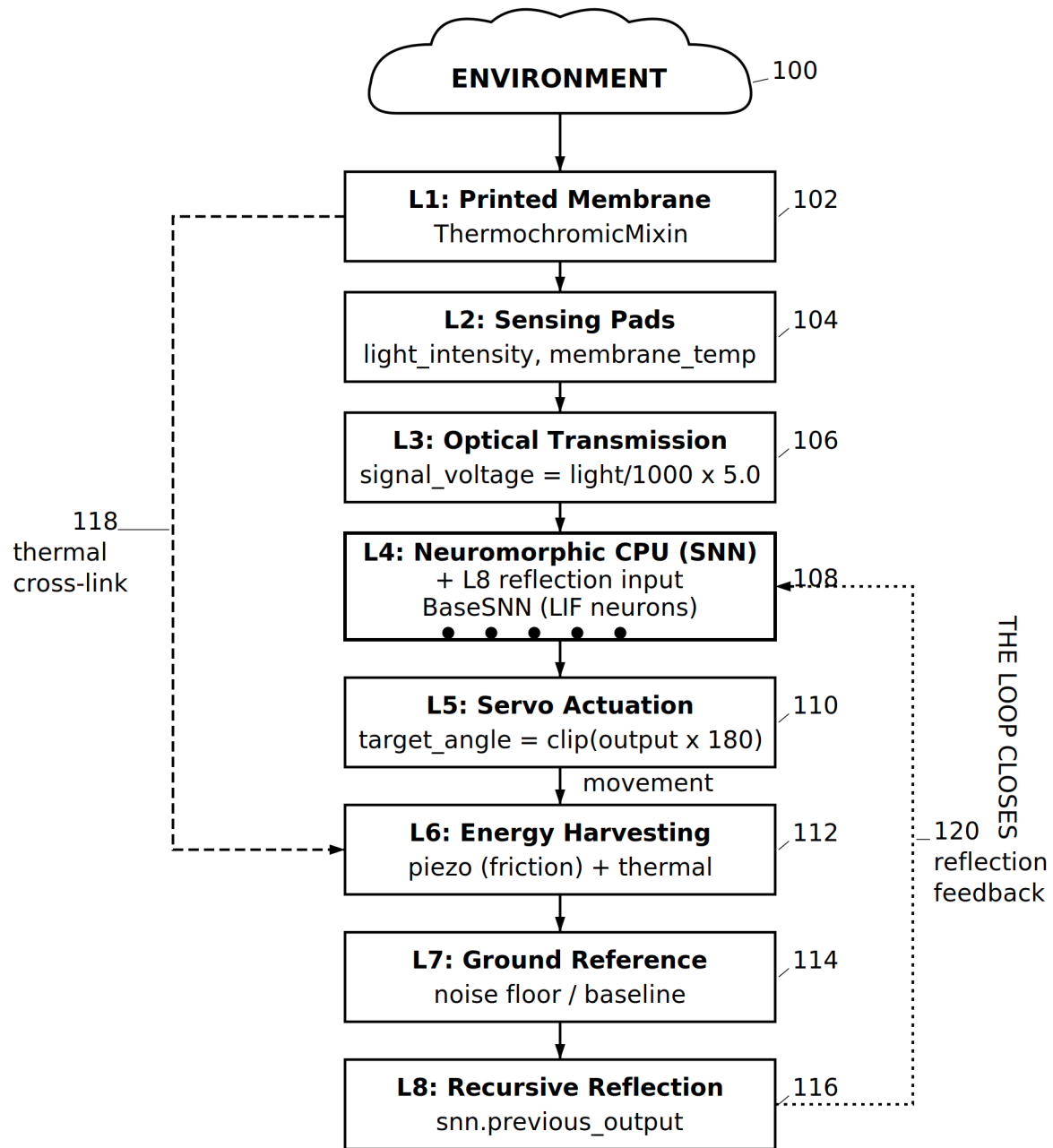
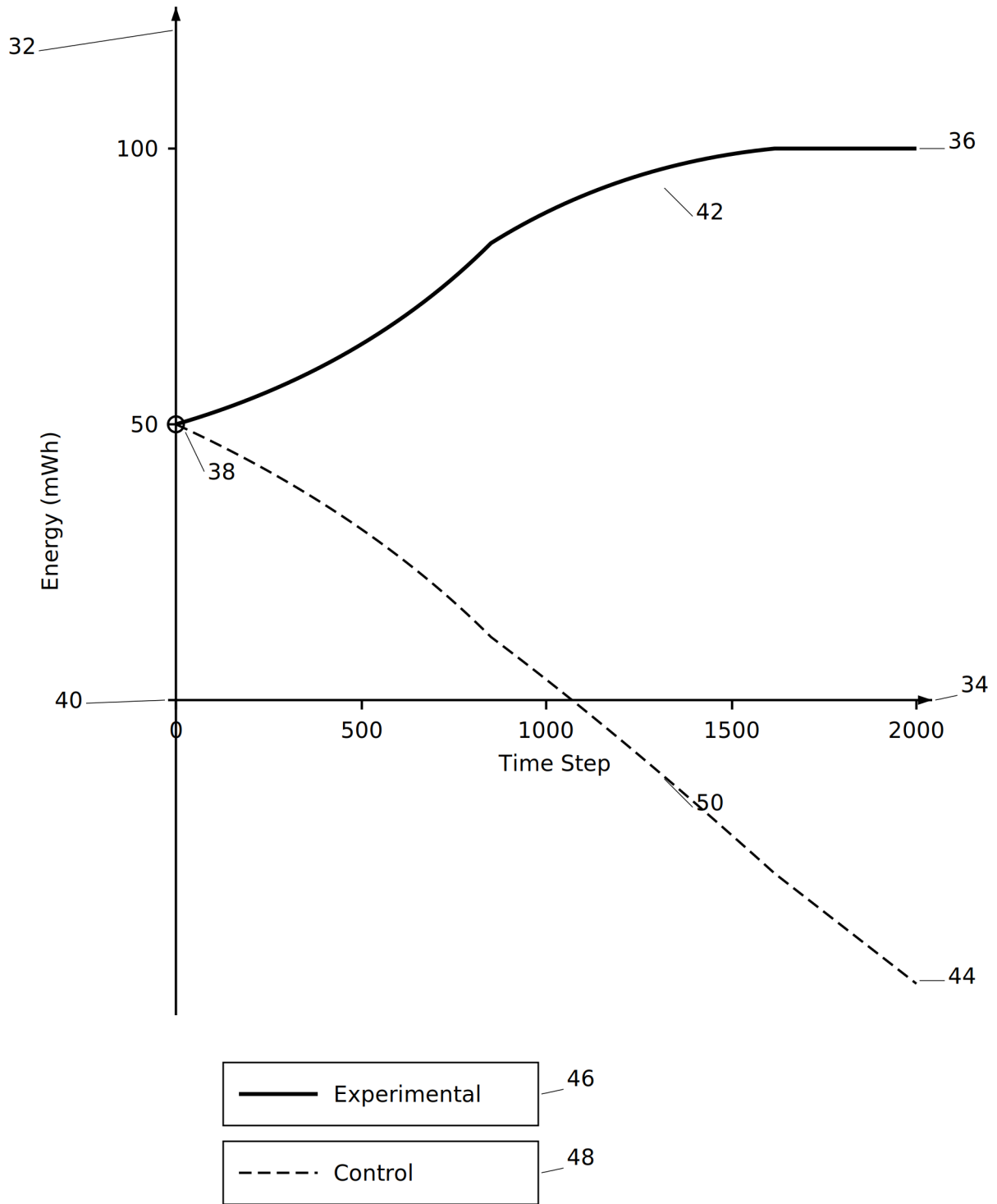


**SIGNAL PATHWAYS:**

- Main spine (L1 through L8 to L4)
- Thermal cross-link (L1 temp to L6)
- ..... Reflection feedback (L8 to L4)

**FIG. 1**



**FIG. 2**

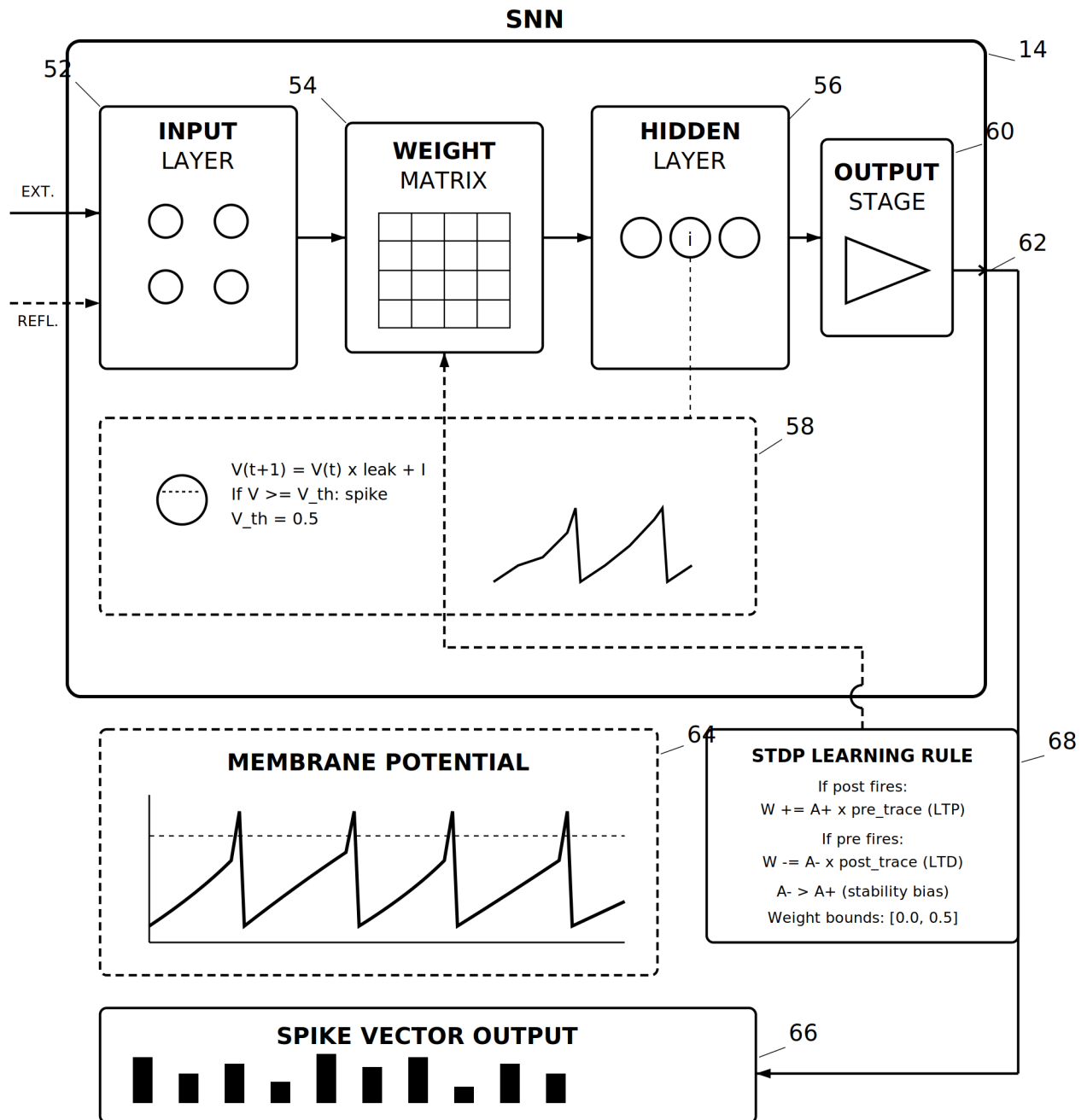


FIG. 3

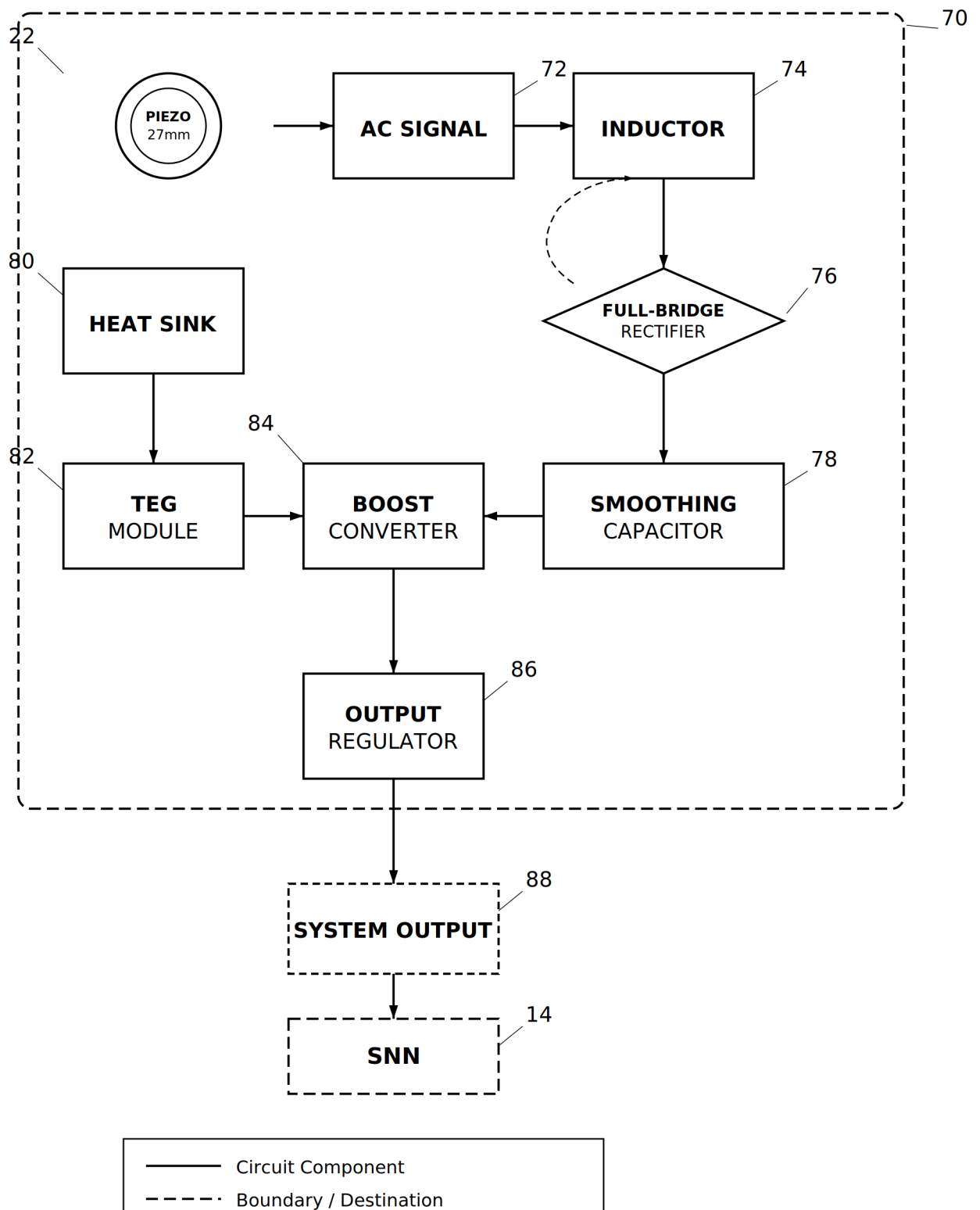


FIG. 4

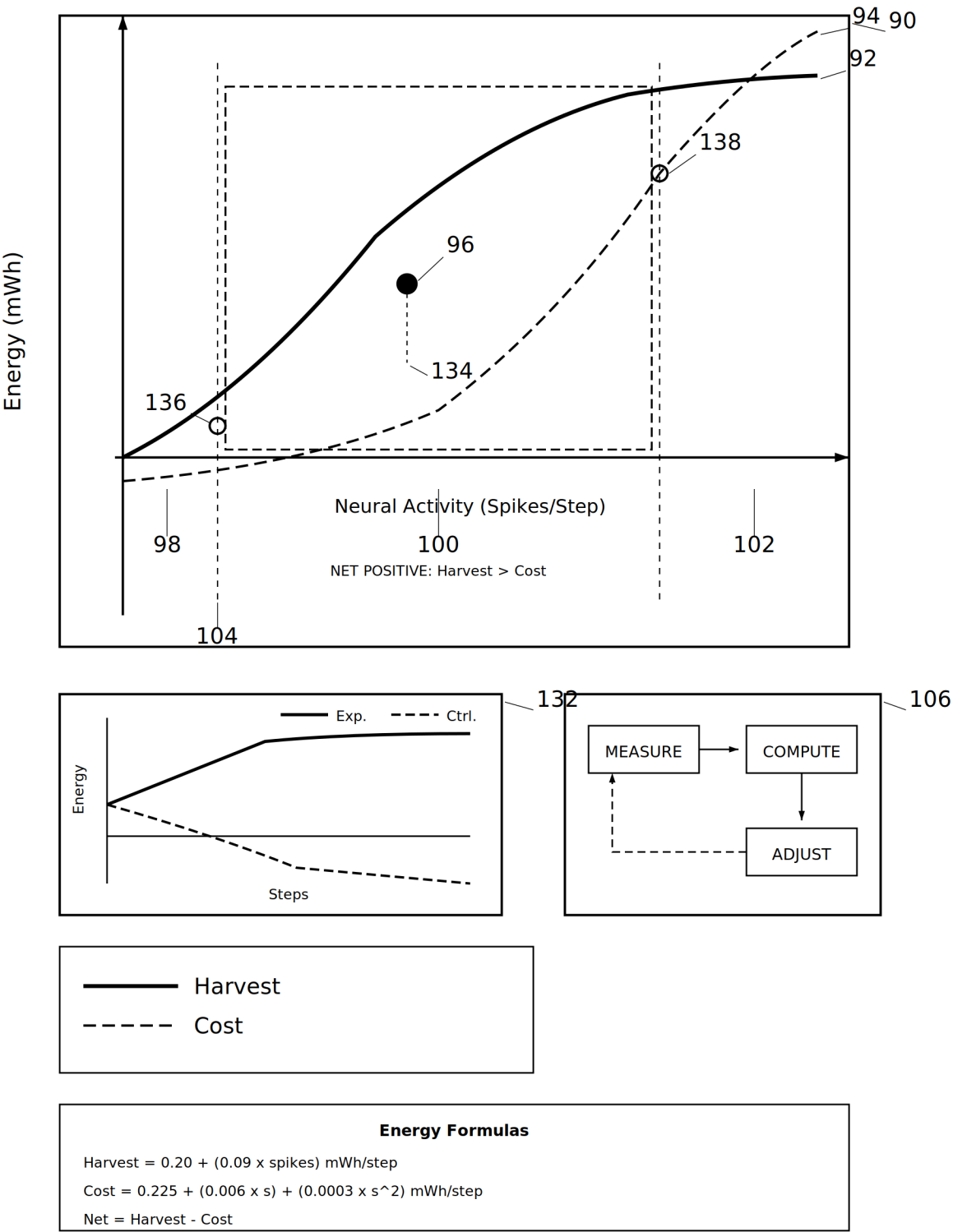
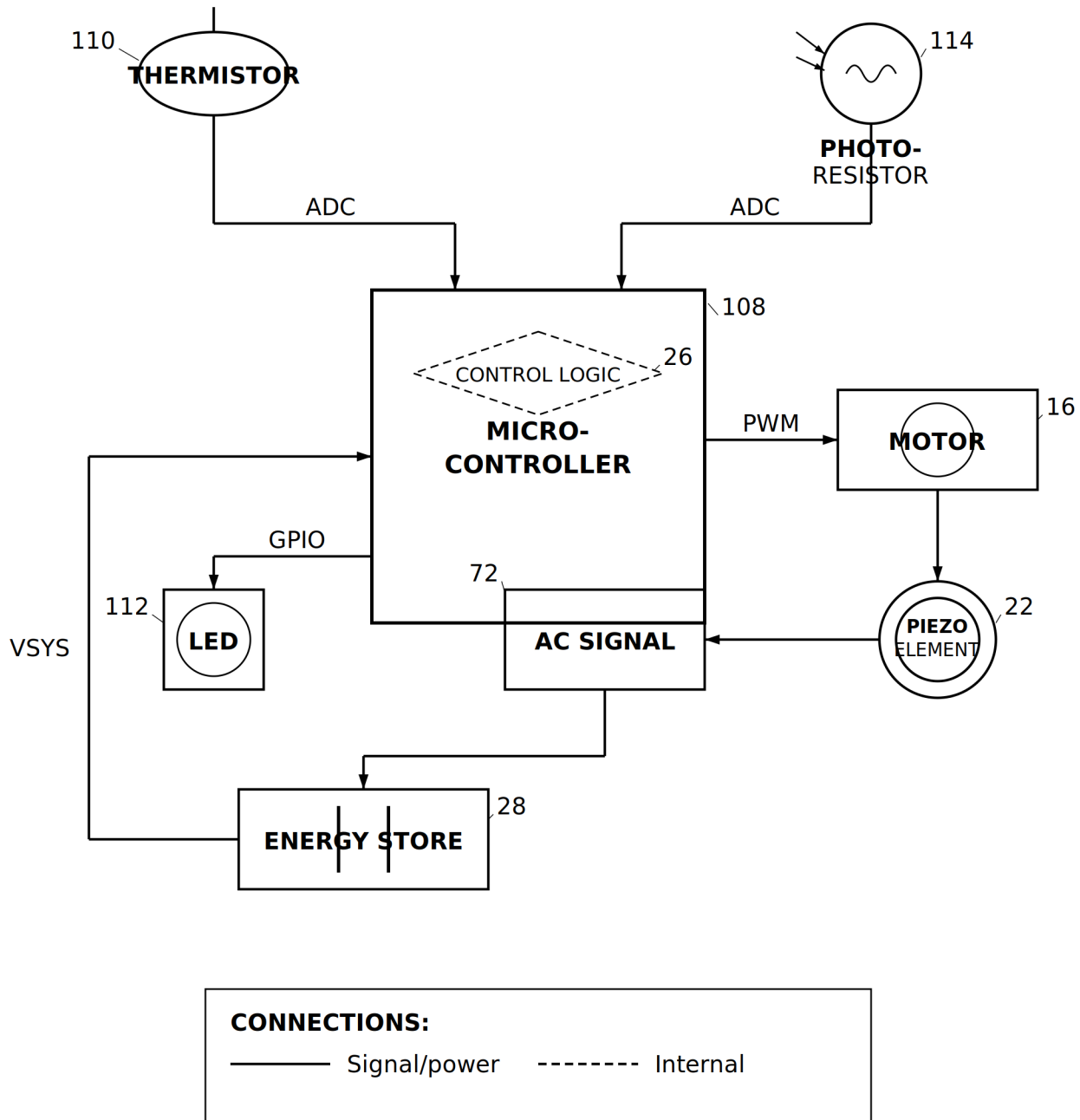
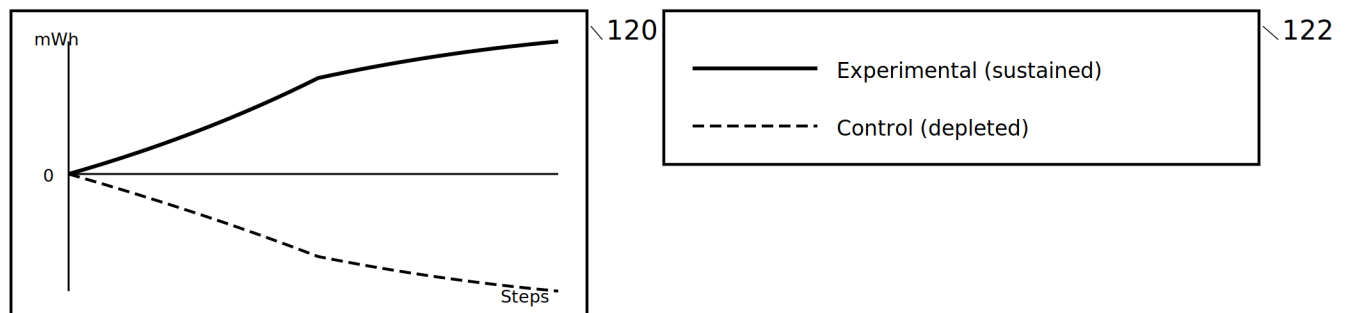


FIG. 5

**FIG. 6**

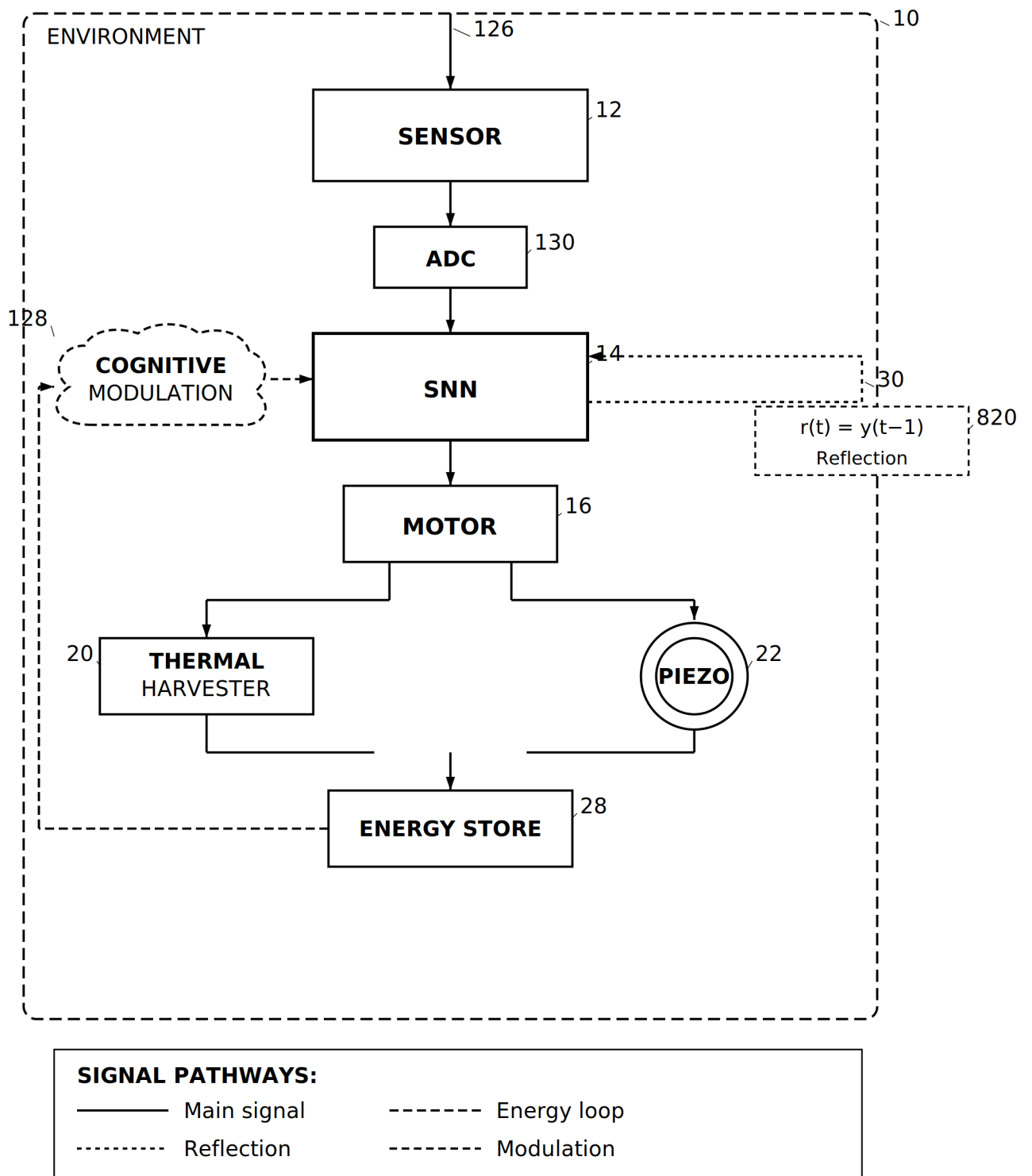
| Claim                                       | Criterion       | Measured      | Threshold        | Result      |
|---------------------------------------------|-----------------|---------------|------------------|-------------|
| <b>Phase 7 : Control Baseline</b>           |                 |               |                  |             |
| A                                           | Continuity      | 2000 pts      | 2000 pts         | <b>PASS</b> |
| B                                           | Boundedness     | finite        | finite           | <b>PASS</b> |
| C                                           | Robustness      | std = 24.8    | std <= 45        | <b>PASS</b> |
| D                                           | I/O gain        | corr = 0.42   | corr >= 0.2      | <b>PASS</b> |
| E                                           | Saturation      | 0.03          | <= 0.20          | <b>PASS</b> |
| <b>Phase 8 : Synaptic Plasticity (STDP)</b> |                 |               |                  |             |
| F8.1                                        | Weight entropy  | 0.94 bit drop | decrease         | <b>PASS</b> |
| F8.2                                        | STDP MI ratio   | 6.52x         | > 1.20x          | <b>PASS</b> |
| F8.3                                        | Convergence     | 0.00%         | < 10%            | <b>PASS</b> |
| <b>Phase 9 : Predictive Processing</b>      |                 |               |                  |             |
| F9.1                                        | Error reduction | 63.9%         | > 50%            | <b>PASS</b> |
| F9.2                                        | Random trend    | p = 0.69      | p > 0.05         | <b>PASS</b> |
| F9.3                                        | Adaptation      | 1.51x spike   | > 1.3x + recover | <b>PASS</b> |
| <b>Phase 10 : Multi-Modal Integration</b>   |                 |               |                  |             |
| F10.1                                       | Sync difference | 0.70          | > 0.20           | <b>PASS</b> |
| F10.2                                       | Weight entropy  | 2.31 to 0.0   | decrease         | <b>PASS</b> |
| F10.3                                       | Convergence     | 0.18          | < 0.20           | <b>PASS</b> |

| Configuration                 | Energy at 2000 Steps | Status    |
|-------------------------------|----------------------|-----------|
| Control (EnergyConfig)        | -4,697 mWh           | Depleted  |
| Experimental (BalancedConfig) | 100.0 mWh            | Sustained |



**14 / 14 CLAIMS PASS : Phases 7 - 10 Validated**

**FIG. 7**

**FIG. 8**